

Enterprise Agentic AI School Transportation Compliance Platform

Abu Dhabi Mobility & ADEK Regulatory Intelligence Demo

Version 1.0

1. Executive Summary

This project demonstrates a next-generation Agentic AI platform for Abu Dhabi school transportation compliance and safety management.

The platform combines:

- Multi-Agent AI Architecture
- LangGraph Orchestration
- MCP (Model Context Protocol)
- A2A (Agent-to-Agent Communication)
- Enterprise RAG
- Real-time Operational Dashboard
- Regulatory Intelligence

The solution simulates a real deployment for Abu Dhabi school transportation operators while using mostly free and open-source technologies.

2. Objectives

Demonstrate:

- Autonomous decision making
 - Multi-agent collaboration
 - Regulatory compliance validation
 - Incident management
 - Executive reporting
 - Knowledge retrieval from official Abu Dhabi regulations
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3. High Level Architecture

User Dashboard ↓

API Gateway (FastAPI) ↓

LangGraph Supervisor Agent ↓

A2A Communication Layer ↓

Specialized Agents

- Compliance Agent
- Safety Agent
- Incident Agent
- Route Optimization Agent
- Executive Insights Agent

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Shared RAG Layer

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MCP Tool Layer

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Mock Enterprise Systems

4. Technology Stack

Frontend

React Next.js Tailwind CSS ShadCN UI Recharts

Backend

Python FastAPI

Agent Framework

LangGraph LangChain

LLM

OpenAI GPT or Llama 3.3 70B (Groq)

Vector Database

ChromaDB

Database

PostgreSQL

Cache

Redis

Observability

LangSmith (Free Tier)

5. Agent Design

Supervisor Agent

Responsibilities:

- Receive user requests
- Plan workflow
- Select agents
- Manage execution state
- Aggregate results

Example:

User: "Why did compliance score decrease?"

Supervisor:

1. Ask Compliance Agent
2. Ask Incident Agent
3. Ask Executive Agent
4. Merge results

Compliance Agent

Responsibilities:

- Driver validation
- Vehicle validation
- Permit verification
- Training verification
- Policy lookup

Tools:

- PASS MCP
- Policy RAG
- Fleet Database

Outputs:

- Compliance decision
- Violation report
- Recommended action

Safety Agent

Responsibilities:

- Driver distraction
- Seatbelt violations
- Student illness
- Speed violations

Tools:

- Incident MCP
- Policy RAG

Outputs:

- Risk score
 - Escalation recommendation
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Incident Agent

Responsibilities:

- Incident creation
- Notification workflows
- Parent alerts
- Emergency escalation

Tools:

- Notification MCP
- Incident Database

Outputs:

- Incident tickets
 - Escalation actions
-

Executive Insights Agent

Responsibilities:

- KPI analysis
- Trend detection
- Compliance analytics
- Root cause analysis

Tools:

- Analytics MCP
- Data Warehouse

Outputs:

- Executive summaries
 - Dashboards
 - Recommendations
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6. A2A Communication Design

Example Workflow

Safety Agent detects:

Driver using mobile phone

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Safety Agent

asks

Compliance Agent

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Check previous violations

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Compliance Agent returns:

3 previous violations

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Safety Agent escalates severity

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Incident Agent creates ticket

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Executive Agent updates risk score

This demonstrates real agent collaboration.

7. MCP Architecture

MCP provides enterprise integrations.

MCP Servers

Policy MCP

Provides:

- ADEK policies
- Abu Dhabi Mobility regulations
- Violation matrix

Fleet MCP

Provides:

- Vehicles
- GPS
- Maintenance records

Driver MCP

Provides:

- Driver records
- Certifications
- Medical clearances

Notification MCP

Provides:

- Email
- SMS
- Push notifications

Incident MCP

Provides:

- Incident creation
- Incident search
- Escalation workflows

8. Knowledge Base Architecture

Knowledge Sources

Folder Structure

knowledge_base/

regulations/

- adek_transport_policy.pdf
- adek_student_protection.pdf
- adek_behaviour_policy.pdf

mobility/

- school_transport_rules.pdf
- route_guidelines.pdf
- licensing_rules.pdf

operations/

- emergency_sop.pdf
- evacuation_sop.pdf
- supervisor_sop.pdf

penalties/

- violation_matrix.pdf
 - fines.pdf
-

9. Vector Database Structure

Chunk Size:

1000 characters

Overlap:

200 characters

Metadata:

{ source document_name section regulation_type authority effective_date }

Example:

```
{ authority: "ADEK", document: "Transportation Policy", section: "Guardian Handover" }
```

10. Mock Data Design

Drivers

500 records

Fields:

driver_id name permit_status medical_status training_status operator

Vehicles

250 records

Fields:

vehicle_id license_plate age gps_status inspection_status

Students

5000 records

Fields:

student_id school route guardian behavior_stage

Incidents

1000 records

Fields:

incident_id severity type driver vehicle timestamp

11. Dashboard Pages

Command Center

Shows:

- Active buses
 - Open incidents
 - Compliance score
 - Safety score
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Fleet View

Map Vehicle status Maintenance status

Compliance Center

Drivers Vehicles Violations Permits

Incident Center

Incident queue Severity analysis Response status

Executive Insights

Monthly compliance trends Risk analysis Recommendations

12. Demo Scenarios

Scenario 1

Driver Mobile Usage

Agent Flow:

Safety Agent → Compliance Agent → Incident Agent → Executive Agent

Result:

Training assigned Violation logged

Scenario 2

Missing Guardian

Agent Flow:

Safety Agent → Policy RAG → Incident Agent

Result:

Student retained on bus Parent notified

Scenario 3

Vehicle Inspection Failure

Agent Flow:

Compliance Agent → Fleet MCP

Result:

Vehicle blocked Maintenance ticket generated

Scenario 4

Executive Question

User:

"Why did compliance decrease in March?"

Agent Flow:

Supervisor → Compliance Agent → Incident Agent → Executive Agent

Result:

13. Free Infrastructure

Frontend

Vercel Free

Backend

Railway Free or Render Free

Vector DB

ChromaDB

Database

Supabase Free

Cache

Redis Cloud Free

LLM

Groq Free Tier Llama 3.3 70B

Monitoring

LangSmith Free

Storage

GitHub

14. Success Metrics

Compliance Accuracy > 95%

Policy Retrieval Accuracy > 90%

Agent Collaboration Success > 95%

Incident Routing Accuracy > 98%

Dashboard Response Time < 2 seconds

RAG Retrieval Time < 1 second

15. Future Enhancements

Real Asateel Integration

Real PASS Integration

Computer Vision Edge AI

Real-time GPS Tracking

Predictive Risk Scoring

Autonomous Compliance Audits

Digital Twin Fleet Simulation